Python Pandas for AI & Robotics

Assignment Questions

Name: _____ Batch / ID: _____ Date: _____

Total: 100 marks Questions: 20 (5 marks each) Time: 90 minutes

- Write Python code with Pandas for every question. Start each program with `import pandas as pd`.
- Where a question says "print", show the statement you would use and the output you expect.
- All scenarios are drawn from robotics and AI. Questions marked (CSV) use the practice file `sensor_data_100x5.csv`: 100 sensor readings with timestamp, temperature_c, humidity_pct, pressure_hpa and vibration_g, including missing values.

Section A: Getting Started with Pandas

Q1. Pandas Intro and Getting Started [5 marks]

A robotics team wants to analyse sensor logs in tables instead of plain lists.

(a) Give two reasons why Pandas is useful for working with sensor data.

(b) Write the statement that imports Pandas with the alias `pd` and prints the installed version.

(c) In one sentence, how is a DataFrame different from a NumPy array?

Q2. Pandas Series [5 marks]

A robot reports three battery readings: 85, 78 and 70 (percent).

(a) Create a Series from these values.

(b) Create the same Series with the labels `"t1"`, `"t2"` and `"t3"` as its index, then print the value at label `"t2"`.

(c) Create a Series from the dictionary `{"lidar": 12, "camera": 30, "imu": 100}` (sensor name : update rate in Hz).

Q3. Pandas DataFrames [5 marks]

Fleet information is given below.

```
data = {"robot": ["Rover-1", "Rover-2", "Drone-1"],
        "battery": [85, 40, 62],
        "speed": [1.2, 0.8, 5.5]}
```

(a) Create a DataFrame named `fleet` from this dictionary.

(b) Print its shape and the names of its columns.

(c) Select only the `"battery"` column, then locate the first row with `loc[0]`.

Q4. Read CSV (CSV) [5 marks]

Load the practice file into a DataFrame.

(a) Read `sensor_data_100x5.csv` into a variable named `df`.

(b) Print the first 5 rows and the last 3 rows.

(c) Read the file again, this time parsing the `timestamp` column as dates.

(d) Print the total number of rows with `len(df)` and explain what `to_string()` would do.

Q5. Read JSON [5 marks]

A robot saves a mission log as JSON.

```
[{"step": 1, "action": "move", "distance": 2.5},
 {"step": 2, "action": "turn", "distance": 0.0},
 {"step": 3, "action": "move", "distance": 4.1}]
```

(a) Read the JSON data below into a DataFrame (assume it is saved as `mission.json`).

(b) Print the DataFrame and its column names.

(c) Name one situation where JSON is a better format than CSV.

Q6. Analyzing Data (CSV) [5 marks]

Explore the sensor file before using it.

(a) Use `head()` with an argument to print the first 10 rows.

(b) Use `info()` and describe what it tells you about data types and missing values.

(c) Use `describe()` to get the mean and standard deviation of every numeric column.

(d) Print the shape of the DataFrame.

Section B: Slicing and Editing Columns

Q7. DataFrame Slicing [5 marks]

Use the `fleet` DataFrame from Q3.

(a) Select the first two rows using a slice.

(b) Select the columns `robot` and `battery` together.

(c) Select the rows where `battery` is below 50.

Q8. Slicing with iloc[] [5 marks]

Use the `fleet` DataFrame from Q3.

(a) Select the row at position 1 using `iloc`.

(b) Select rows 0 to 1 and columns 0 to 1 with a single `iloc` expression.

(c) Select the last row using a negative position.

(d) Explain how `iloc` differs from `loc`.

Q9. Slicing with loc[] [5 marks]

Here the DataFrame uses robot names as its index.

```
df = pd.DataFrame({"battery": [85, 40, 62],
                   "speed": [1.2, 0.8, 5.5]},
                  index=["Rover-1", "Rover-2", "Drone-1"])
```

(a) Create the DataFrame below.

(b) Select the row for `"Rover-2"` using `loc`.

(c) Select the `battery` of `"Drone-1"` and `"Rover-1"` in one expression.

(d) Select all rows with a `speed` above 1.0 and show only the `battery` column.

Q10. Renaming Columns and Rows [5 marks]

Rename parts of the DataFrame from Q9.

(a) Rename the column `battery` to `battery_pct` using `rename()`.

(b) Rename the row label `"Drone-1"` to `"UAV-1"`.

(c) Replace all column names at once by assigning a new list to `df.columns`.

Q11. Adding or Inserting Columns [5 marks]

Use the `fleet` DataFrame from Q3.

(a) Add a new column `speed_kmph` equal to `speed` multiplied by 3.6.

(b) Add a column `status` with the value `"active"` for every row.

(c) Insert a column `type` at position 1 using `insert()`.

Q12. Updating Column Values [5 marks]

Use the `fleet` DataFrame from Q3.

(a) Set the battery of `"Rover-2"` to 100 after charging, using `loc`.

(b) Increase every value in the `speed` column by 10 percent.

(c) Replace `battery` values below 50 with the text `"low"` using a condition.

Q13. Removing Unnecessary Columns [5 marks]

A sensor file has an unused column named `"notes"`.

(a) Remove the column with `drop()` and the correct `axis` or `columns` argument.

(b) Remove it in place without creating a new DataFrame.

(c) Remove the two columns `"notes"` and `"id"` in one call.

(d) Use `del` to remove a column and state one difference from `drop()`.

Section C: Cleaning Data

Q14. Cleaning Data [5 marks]

Raw sensor files are rarely ready for analysis.

(a) List the four kinds of bad data covered in this course (empty cells, wrong format, wrong data and duplicates).

(b) For each kind, give one robotics example (for example a failed temperature reading).

(c) Why must data be cleaned before it is used to train an AI model?

Q15. Cleaning Empty Cells (CSV) [5 marks]

The practice file has missing values in four columns.

(a) Count the missing values in each column using `isna()` and `sum()`.

(b) Create a new DataFrame with all rows containing empty cells removed using `dropna()`. How many rows remain?

(c) Fill missing `temperature_c` values with the column mean using `fillna()`.

(d) Fill missing `humidity_pct` values with the median instead, and explain when the median is a better choice.

Q16. Cleaning Wrong Format [5 marks]

A log has dates and numbers stored as text.

```
df = pd.DataFrame({"date": ["2026-10-14", "2026-10-15", "2026/10/16"],
                   "distance": ["2.5", "4.1", "3.8"]})
```

(a) Convert the `date` column to real dates using `pd.to_datetime()`.

(b) Convert the `distance` column to numbers using `pd.to_numeric()`.

(c) Check the data types again with `dtypes` to confirm the change.

Q17. Cleaning Wrong Data [5 marks]

A motor temperature column contains impossible readings.

```
df = pd.DataFrame({"motor": ["A", "B", "C", "D"],
                   "temp": [65, 72, 950, 68]})
```

(a) Replace every value above 120 in `temp` with 120 using a condition and `loc`.

(b) Loop over the rows and drop any row whose `temp` is above 120.

(c) Explain why wrong values like a battery of 250 percent should be fixed or removed.

Q18. Removing Duplicates [5 marks]

A sensor logger accidentally wrote some rows twice.

```
df = pd.DataFrame({"time": [1, 2, 2, 3, 3],
                   "reading": [10, 12, 12, 11, 11]})
```

(a) Use `duplicated()` to find which rows are repeats.

(b) Remove all duplicate rows with `drop_duplicates()`.

(c) Print how many rows were removed.

Section D: Analysis and Visualisation

Q19. Pandas Correlations (CSV) [5 marks]

Does humidity depend on temperature in the sensor data?

(a) Use `corr()` to find the correlation between `temperature_c` and `humidity_pct`.

(b) Print the full correlation table of the DataFrame.

(c) Explain what a correlation near -1, near 0 and near 1 would mean.

(d) Name two columns you would expect to be strongly related, and say why.

Q20. Pandas Plotting (CSV) [5 marks]

Visualise the sensor data with Pandas and Matplotlib.

(a) Import `matplotlib.pyplot` and draw a line plot of `temperature_c`.

(b) Draw a scatter plot of `temperature_c` (x) against `humidity_pct` (y).

(c) Draw a histogram of `vibration_g` and add a title.

(d) Show the plot with `plt.show()`.